Feature	Rationale
FREQUENCY_30Dprev	Captures customer transaction activity in the previous 30-day period. Helps identify whether engagement is increasing, decreasing, or remaining stable over time.
FREQUENCY_30D / FREQUENCY_30Dprev	Measures the relative change in transaction frequency between consecutive 30-day windows. Values above 1 indicate growing activity, while values below 1 suggest declining engagement. Useful for detecting behavioral shifts that may precede churn or increased usage.
FREQUENCY_90D	Represents transaction activity over a longer 90-day horizon. Provides a more stable view of customer behavior by smoothing short-term fluctuations. Helps distinguish consistently active customers from temporarily active ones.
ACTIVE_DAYS_30D	Counts the number of distinct days with transactions during the last 30 days. Reflects engagement consistency rather than transaction volume alone. Customers active on more days generally exhibit stronger platform dependence and loyalty.
TENURE	Measures the duration of the customer's relationship with the institution. Long-tenured customers often display different transaction patterns and risk profiles than newer customers. Helps capture lifecycle effects in customer behavior.
SERVICE_DIVERSITY	Indicates the variety of banking services or transaction types used by the customer. Higher diversity typically reflects deeper product adoption and stronger engagement. Customers using multiple services are generally less likely to become inactive.
AVG_TRX_GAP_DAYS	Measures the average number of days between consecutive transactions. Smaller gaps indicate frequent engagement, while larger gaps may signal declining activity. Helps quantify transaction regularity.
AVGBALANCE30D	Represents the customer's average account balance during the most recent 30 days. Captures current financial capacity and

Feature	Rationale
	account utilization. Higher balances may indicate greater relationship value and engagement.
AVGBALANCE30Dprev	Measures the average balance in the preceding 30-day period. Provides a historical benchmark for evaluating balance growth or decline. Useful for identifying changes in financial behavior.
AVGBALANCE30D / AVGBALANCE30Dprev	Captures the relative change in average balance between two consecutive periods. Rising balances may indicate increasing account usage or wealth accumulation, while declining balances may signal disengagement or fund withdrawals.
RECENCY	Measures the number of days since the customer's most recent transaction. One of the strongest indicators of engagement, as inactive customers typically exhibit high recency values. Frequently used in RFM-based behavioral modeling.
FREQUENCY	Represents the total number of transactions during the observation period. Reflects overall customer activity and engagement level. Higher frequency generally indicates stronger product usage.
FREQUENCY_30D	Captures transaction activity in the most recent 30-day period. Provides a timely view of customer behavior and responsiveness. Particularly useful for detecting recent changes in engagement.
MONETARY	Measures the total value of transactions within the observation window. Captures the economic significance of customer activity beyond transaction counts. High monetary value often indicates more valuable customer relationships.
BPR	Represents the balance preservation ratio, measuring how much balance is retained relative to transaction activity. Helps distinguish customers who maintain funds from those who frequently deplete balances. Can indicate financial stability and account dependence.

Feature	Rationale
BALANCE_VOLATILITY	Quantifies fluctuations in account balance over time. High volatility may reflect irregular financial behavior or unstable cash flow patterns. Stable balances often indicate predictable account usage.
TRX_AMT_VOLATILITY	Measures variation in transaction amount